

PROJECT REQUIREMENT DOCUMENT

Namma-Raste Reporter (Infrastructure)

1. Introduction

Namma-Raste Reporter is an Android application designed to simplify reporting of road infrastructure issues such as potholes and broken streetlights.

2. Problem Statement

Potholes and damaged streetlights are major causes of road accidents. Current reporting methods are slow and inefficient.

3. Objective

- 1 Provide one-click reporting system
- 2 Capture real-time image and GPS location
- 3 Generate unique Ticket ID
- 4 Improve communication with authorities

4. Scope of Project

- 1 Users can report anytime
- 2 Works offline with local storage
- 3 Usable in rural and urban areas

5. Features

- 1 Capture image using camera
- 2 Select issue type
- 3 Auto GPS location
- 4 Generate Ticket ID
- 5 Store data locally
- 6 View previous reports

6. Technology Stack

- 1 Kotlin
- 2 XML / Jetpack Compose
- 3 Room Database
- 4 CameraX API
- 5 Fused Location Provider

7. Functional Requirements

- 1 Capture image
- 2 Fetch location
- 3 Generate Ticket ID
- 4 Save and display reports

8. Non-Functional Requirements

- 1 Fast and responsive
- 2 User-friendly UI
- 3 Works on mid-range devices
- 4 Secure local storage

9. Expected Outcome

- 1 Faster reporting
- 2 Improved road safety
- 3 Reduced accidents

10. Future Enhancements

- 1 Firebase integration
- 2 Admin dashboard
- 3 Notifications
- 4 Map view

11. Conclusion

This app provides a simple solution to report infrastructure issues and improves communication between citizens and authorities.